

MODULE 4: Network Management & Physical Layer

PCCST501 Computer Networks | Quick Exam Revision Guide | KTU S5 CSE | Prof. Anju Markose, VJCET

1. Network Management: SNMP & ASN.1

SNMP Architecture: Manager (NMS), Managed Agent, Management Information Base (MIB), and Structure of Management Information (SMI).

- **SNMP PDUs (UDP Port 161 Manager → Agent, Port 162 Agent → Manager):** GetRequest, GetNextRequest, GetBulkRequest, SetRequest, Response, Trap (unsolicited alarm), InformRequest.
- **SNMP Versions:** SNMPv1 (plaintext community string), SNMPv2c (GetBulkRequest), SNMPv3 (USM user security: SHA/MD5 authentication + AES encryption; VACM view access control).
- **ASN.1 (Abstract Syntax Notation One):** Formal language specifying data structures independently of machine architecture.
- **BER (Basic Encoding Rules):** Encodes ASN.1 using TLV (Type-Length-Value) triplet format.
- **MIB Tree & OID:** Hierarchical tree under iso.org.dod.internet.mgmt.mib-2 (1.3.6.1.2.1). E.g. sysDescr = 1.3.6.1.2.1.1.1.

2. Physical Layer: Data, Signals & Capacity Limits

Transmission Impairments: Attenuation (signal energy loss, measured in $\text{dB} = 10 \log_{10}(P_2/P_1)$), Distortion (phase shifts across harmonics), Noise (Thermal, Induced, Crosstalk, Impulse).

Key Concept / Exam Formula:

Theoretical Channel Capacity Formulas:

- **Nyquist Bit Rate (Noiseless Channel):** $C = 2 * B * \log_2(L)$ bps

Where B = Bandwidth in Hz, L = Number of discrete signal levels.

- **Shannon Capacity Theorem (Noisy Channel):** $C = B * \log_2(1 + \text{SNR})$ bps

Where SNR is linear power ratio: $\text{SNR}_{\text{dB}} = 10 * \log_{10}(\text{SNR}_{\text{linear}})$. E.g., 30 dB → SNR = 1000.

3. Digital Transmission: Line Coding Techniques

- **Unipolar NRZ:** 1 = High voltage, 0 = 0V. Simple, but large DC component and clock synchronization loss.
- **Polar NRZ-L vs NRZ-I:** NRZ-L (0 = Positive, 1 = Negative). NRZ-I (Transition at bit boundary for 1, no transition for 0).
- **Manchester Encoding (IEEE 802.3):** 0 = High-to-Low transition, 1 = Low-to-High transition in middle of bit interval. Self-synchronizing, requires 2x bandwidth.
- **Differential Manchester:** Transition always present at mid-bit for clocking; transition at beginning represents 0, no transition at beginning represents 1.
- **Bipolar AMI:** 0 = 0V, 1 = Alternating positive and negative voltages. Zero DC component; scrambled by B8ZS/HDB3 to prevent long runs of 0s.

4. Analog Transmission, Multiplexing & Media

- **Digital-to-Analog Modulation:** ASK (Amplitude Shift Keying), FSK (Frequency Shift Keying), PSK (Phase Shift Keying: BPSK, QPSK), QAM (Quadrature Amplitude Modulation: combines ASK + PSK; 16-QAM carries 4 bits/ baud, 64-QAM carries 6 bits/ baud).

- **Multiplexing:** FDM (Analog frequency bands), WDM (Optical light wavelengths over fiber), Synchronous TDM (Fixed time slots), Statistical TDM (On-demand time slots).
- **Guided Transmission Media:** Twisted Pair (UTP Cat 5e/6, STP), Coaxial Cable (copper conductor + mesh shield), Optical Fiber (Step-Index & Graded-Index, Single-mode vs Multi-mode; immune to EMI, immense bandwidth).
- **Unguided Media:** Radio Waves (Omnidirectional, wall penetrating), Microwaves (Line-of-sight parabolic dish), Infrared (Short-range line of sight).